HALALAN REVAMP

System Requirements

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Client: University Computer Center

https://cs191-ame.github.io/blog/

CS 191 WFR1
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1. Web App Requirements Model

1.1 Content Model

The data tree below shows the content model for the Halalan WebApp, with the main content objects Election, Block, Ballot Receipt, and Administrator. The notation follows that of Pressman's Figure 7.10 (Chapter 7, Section 7.5.4, "Content Model for WebApps"). Data items are drawn as unshaded rectangles, content objects as shaded rectangles, and composite items as lighter shaded rectangles marked by a line near the right edge. The first figure shows the top level, and the figures that follow expand each branch.

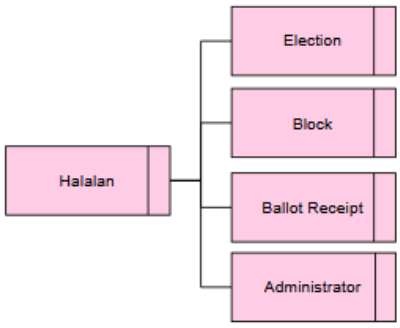


Figure 1. Halalan Data

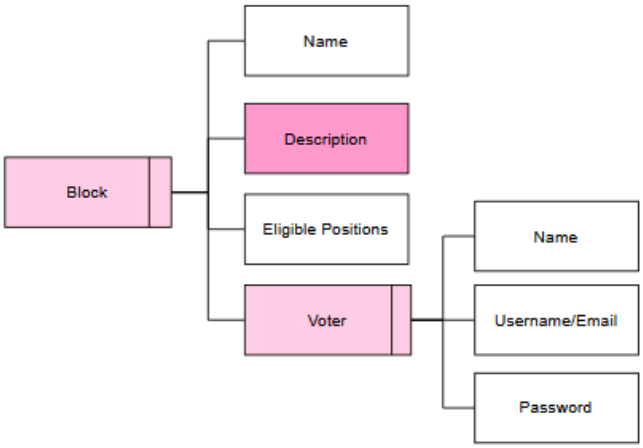


Figure 2. Block Data

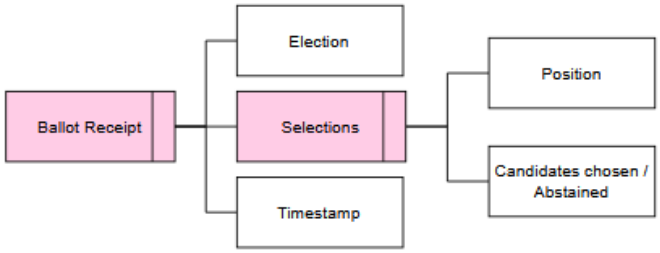


Figure 3. Ballot Receipt Data

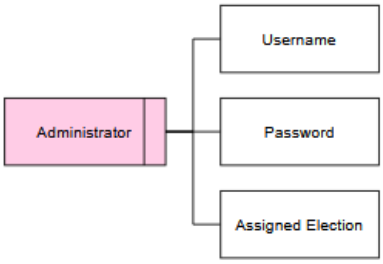


Figure 4. Administrator Data

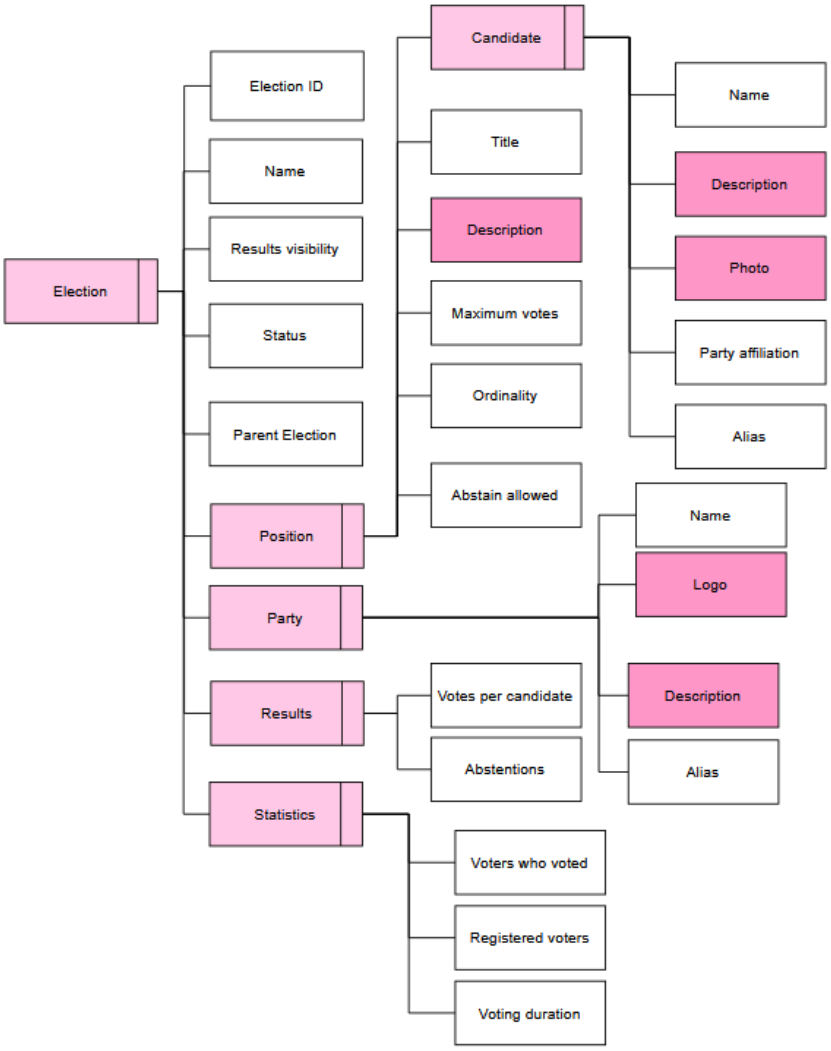


Figure 5. Election Data

1.2 Interaction Model

Actors and Goals

For this election system, 3 main actors have been identified: the administrators (admins), the voters, and external authenticators. The administrators and the voters are primary actors, while the external authenticators are secondary actors.

The admins are individuals in charge of managing the elections. The main admins are personnel from the University of the Philippines Computer Center. They will be responsible for the creation of the elections, for compiling and uploading into the system all relevant, and overseeing the election proper. For the university-wide elections, admin status may be extended to select individuals from each college or department so that they may manage the start and duration of their respective elections. They are considered primary actors as they will be initiating the various events that occur in the system.

The voters consist of members of the university that are eligible and registered to vote in the elections. During the election proper, they are to first, login to the system using their university-issued email address (UP email) and have their identity verified using 2-factor authentication. Then, they may fill out their ballots by selecting candidates. Once they are ready to submit and their vote has been validated by the system, they will be presented with a summary of their vote alongside a prompt to finalize their ballot. Upon submission, they will need to successfully answer a reCAPTCHA for the system to fully process their cast ballot. Voters are a primary actor as well since their actions drive the interactions with the system.

Authenticators such as Google Authenticator and Google reCAPTCHA on the other hand are secondary or participating actors in the system as their services or actions are only triggered as a response to the actions of a primary actor, particularly the voters.

Use Cases

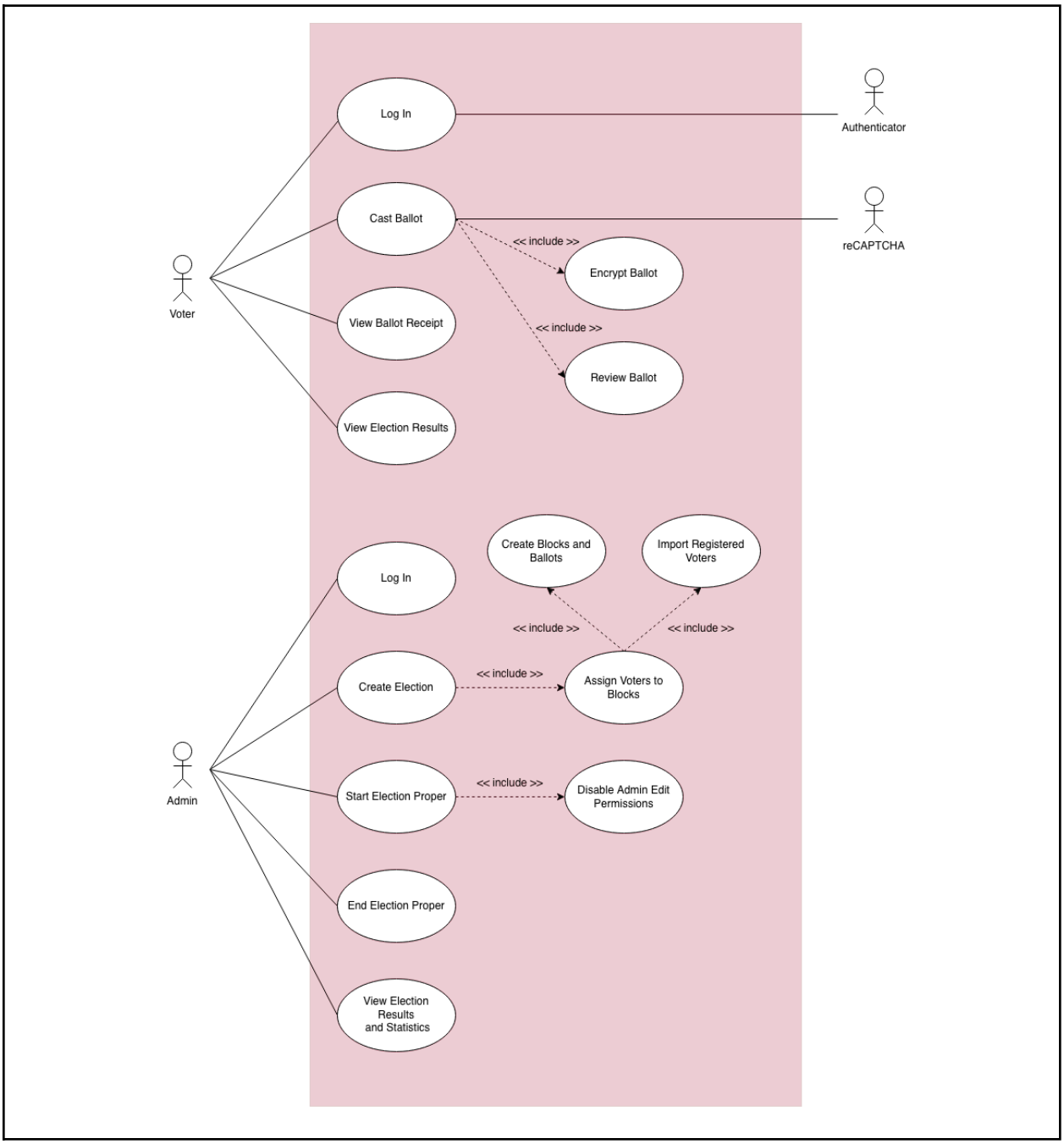


Figure 6. Use Cases Diagram

VOTER.

Log In. A voter can log into the system using their UP email. After inputting their credentials, the system will send a request to the authenticator (Google Authenticator) for 2-factor authentication. Once the voter successfully accomplishes this, they will be logged into the system.

Cast Ballot. During the election proper, a voter can fill out their ballot and cast it. Some subroutines of this case include a display of a summary of their ballot to allow them to review it prior to casting, and the encryption of the ballot, handled by the backend (CodeIgniter). Finally, the submission of the ballot will also send a request for a reCAPTCHA. Only once the voter passes the reCAPTCHA will their vote be stored in the database.

View Ballot Receipt. Following their voting, a voter may log into the site in order to view their ballot receipt containing a summary of their vote.

View Election Results. A voter may track the results of the election in real time by viewing the election results on the site.

ADMIN.

Log In. An admin account may log into the system in order to edit or manage elections.

Create Election. An admin can create an election. Other than inputting the information and parameters for the election, other significant subroutines for the creation of elections are the creation of blocks which dictate the ballot of a certain subset of voters, the importing of the roster of registered voters, and the mapping of these voters to their respective blocks.

Start Election Proper. Once the election is scheduled to start, the admin must publish the election and set its status to “Active”. Afterwards, the system must disable admin permissions to edit any information or data related to the elections that might tamper with the results. This is a subroutine and the election can not proceed without this.

End Election Proper. Once the duration of the election has lapsed, an admin may officially end the election and set its status to “Inactive”. This will finalize all the results.

View Election Results and Statistics. Throughout the election and after it, an admin may view the running results and statistics.

System Sequence Diagrams



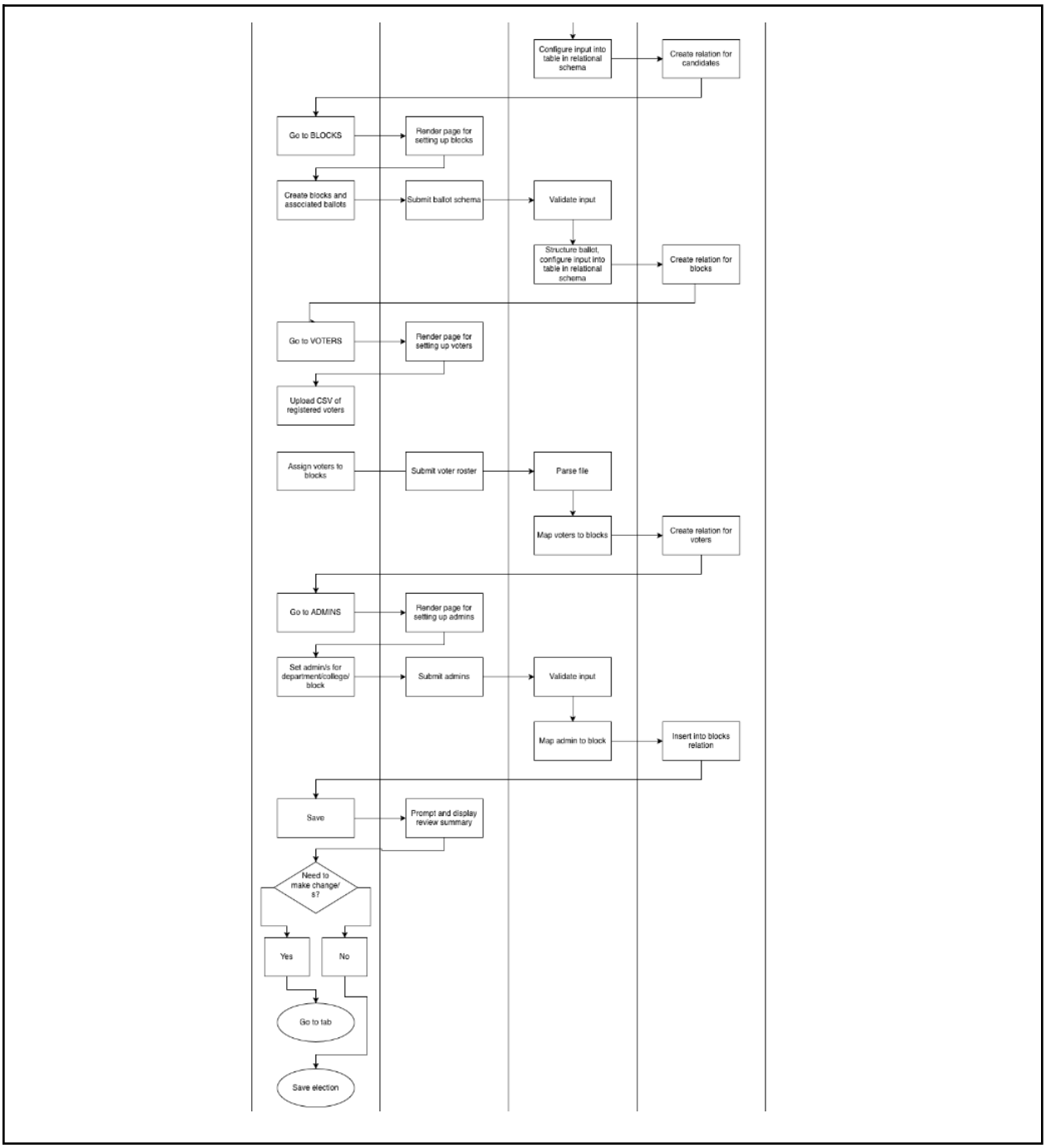


Figure 7. Swimlane Diagram for Admin Use Case “Create Election”

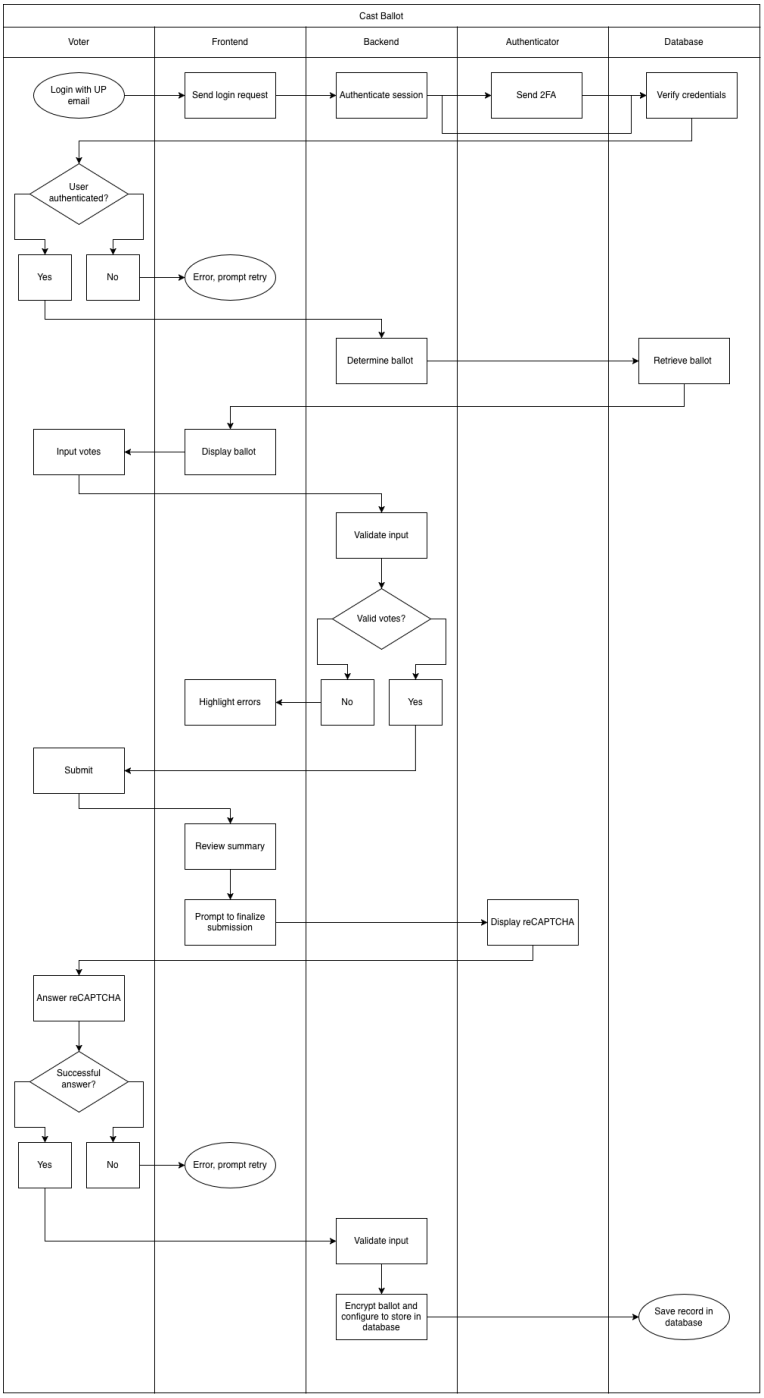


Figure 8. Swimlane Diagram for Voter Use Case “Cast Ballot”

1.3 Configuration Model

This section specifies the allocation of software functionality and subsystems within the physical computing environment.

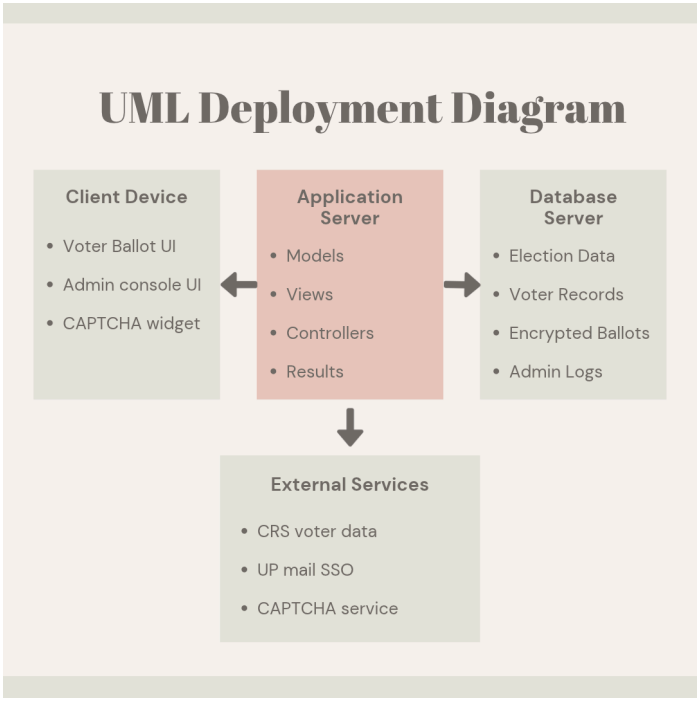


Figure 9. UML Deployment Diagram

The UML Deployment Diagram shown above has three main nodes: the client device, the application server, and the database server, with three external services needed below. The application server and database server can start on one VPS during development. They are separate nodes in the deployment model so the database can be isolated and scaled independently, and it should only accept connections from the application server.

2. Web App Architecture and Design

2.1 Architecture Design

Architecture Genre

Halalan fits the **Government** genre as defined by Grady Booch (cited by Pressman in Section 9.2), which covers systems that support the conduct and operations of a local, state, federal, global, or other political entity. As a voting platform, it is designed to support and conduct the student elections of the University of the Philippines Diliman, which are part of the student government's electoral process.

Architecture Style:

Halalan uses a **layered** architecture style and applies the **Model-View-Controller (MVC)** pattern. The user interface layer contains the views that voters and administrators see. The application layer contains the controllers and models that implement election logic including block-based eligibility, vote and abstain handling, and results, together with authentication and validation services. The data layer holds the MySQL database.

Content Architecture

The **composite** content structure is the most appropriate for Halalan since voters and administrators follow different flows. The voter side uses a **linear structure**, moving from sign-in to choosing candidates, confirming selections and receiving a receipt. The administrator side uses a **hierarchical structure**, in which the content is split into main tabs and further options sit within each tab.

Quality Requirements Tree

Technical Attribute	Sub-attribute	Addressed by
Usability	Global site understandability	A consistent, uncluttered layout for administrators and a single guided path for voters.
	Online feedback and help features	Messages for failed sign-in and invalid input, and a confirmation step before a ballot is submitted.
	Interface and aesthetic features	A shared template for all pages, and a layout that works on most devices.
	Special features	A sample ballot per block and a printable ballot receipt
Functionality	Searching and retrieving capability	Results and statistics filtered by election and block.
	Navigation and browsing features	The sign-in page leads to the area for that user, and admin tabs reach each management section.
	Application domain-related features	Block-based ballot eligibility, abstention, nested elections and per-block results.
Reliability	Correct link processing	Role checks that send each user to their own area. (e.g voter can’t open admin pages)
	Error recovery	Failed sign-in or an expired session returns the user to the Gate with a message.

	User input validation and recovery	The system checks every ballot before saving it, including the maximum votes per position, and rejects invalid ones with a message so the voter can fix them.
Efficiency	Response time performance	Pages are built on the server and sent ready to display, so voters get quick responses.
	Page generation speed	Totals such as the vote counts are calculated directly by the database in one step, instead of being added up one by one.
	Graphics generation speed	The pages use only small images, such as logos and candidate photos, so they load quickly.
Maintainability	Ease of correction	MVC separation. A fix only requires modifying an isolated part of the codebase
	Adaptability	Settings and private details such as passwords and keys are kept in one separate file, so they can be changed without touching the program itself. The sign-in method can also be switched between Google and username and password.
	Extensibility	Elections, positions, parties and blocks are stored as data, so adding a new one doesn’t require changing the program. Changes to the code and

		database structure are documented properly.

Architectural Decisions

Decision	Argument	Alternatives rejected
Rebuild on CodeIgniter(CI) 4	CI 4 is lighter than the alternatives.	Laravel
Layered architecture with MVC	The pattern is already familiar to the team.	Other patterns

2.2 Content (Data) Design

Class Diagrams

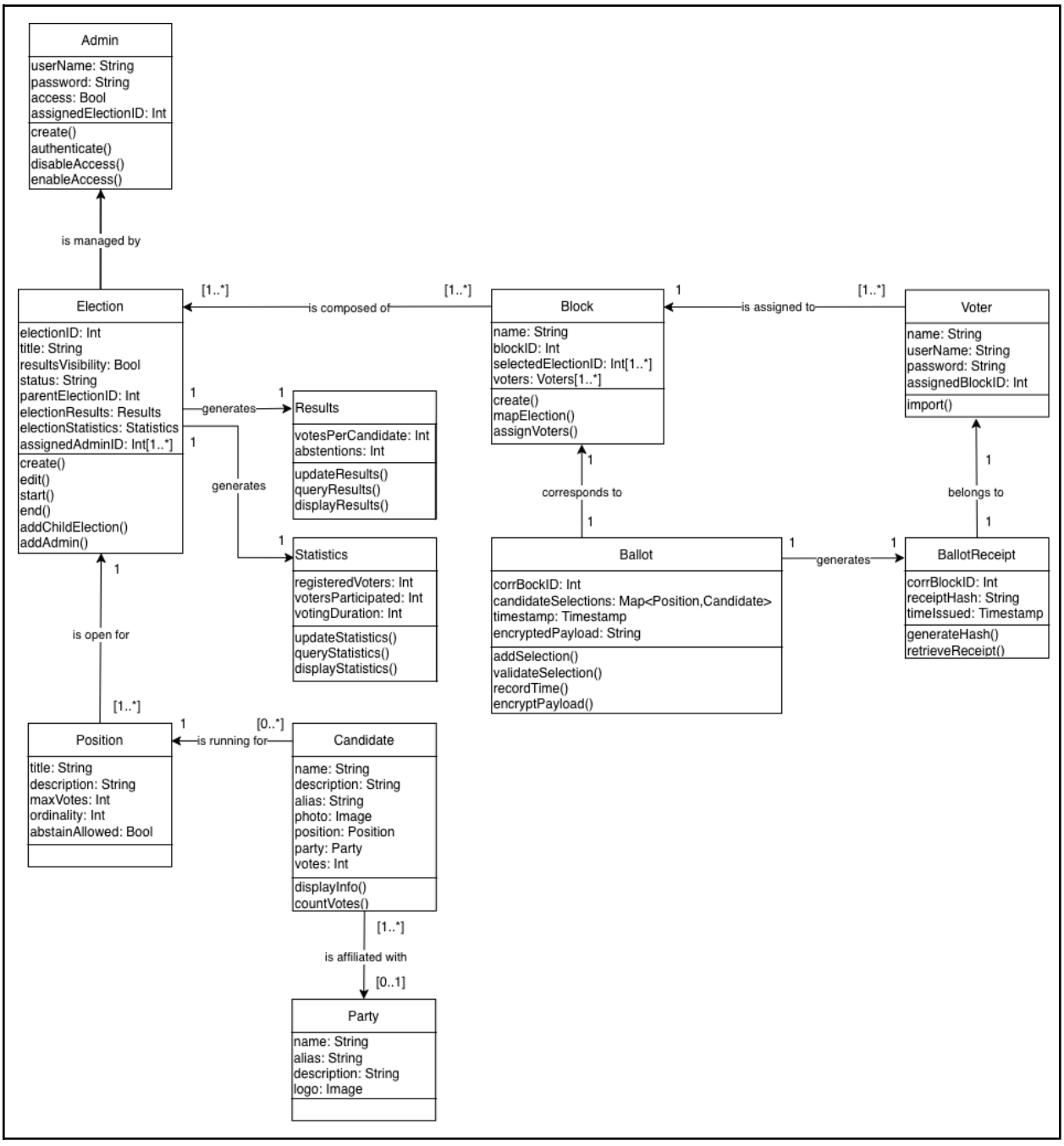


Figure 10. Class Diagram

Entity-Relationship Diagrams

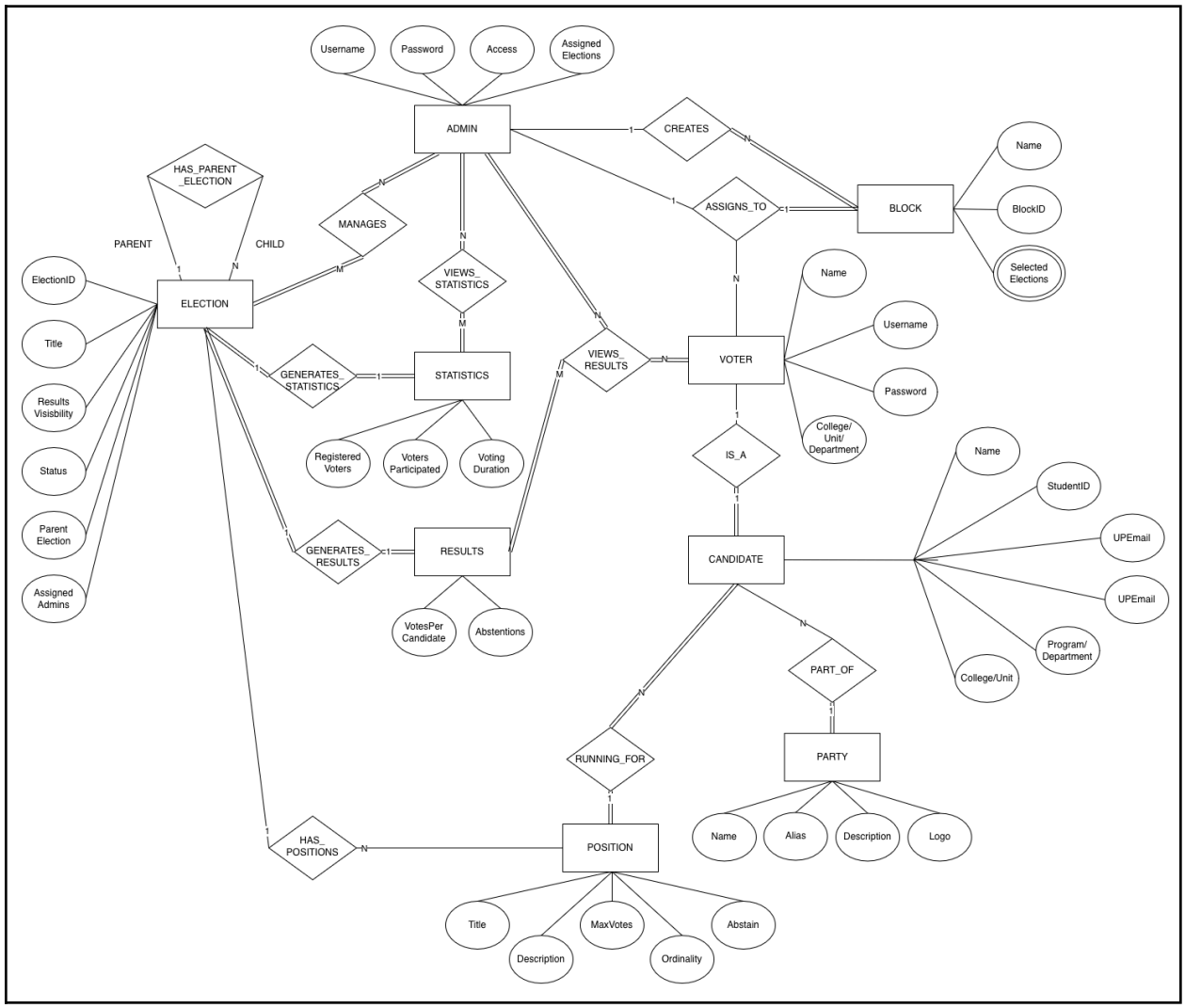


Figure 11. ER Diagram

Data Flow Diagrams

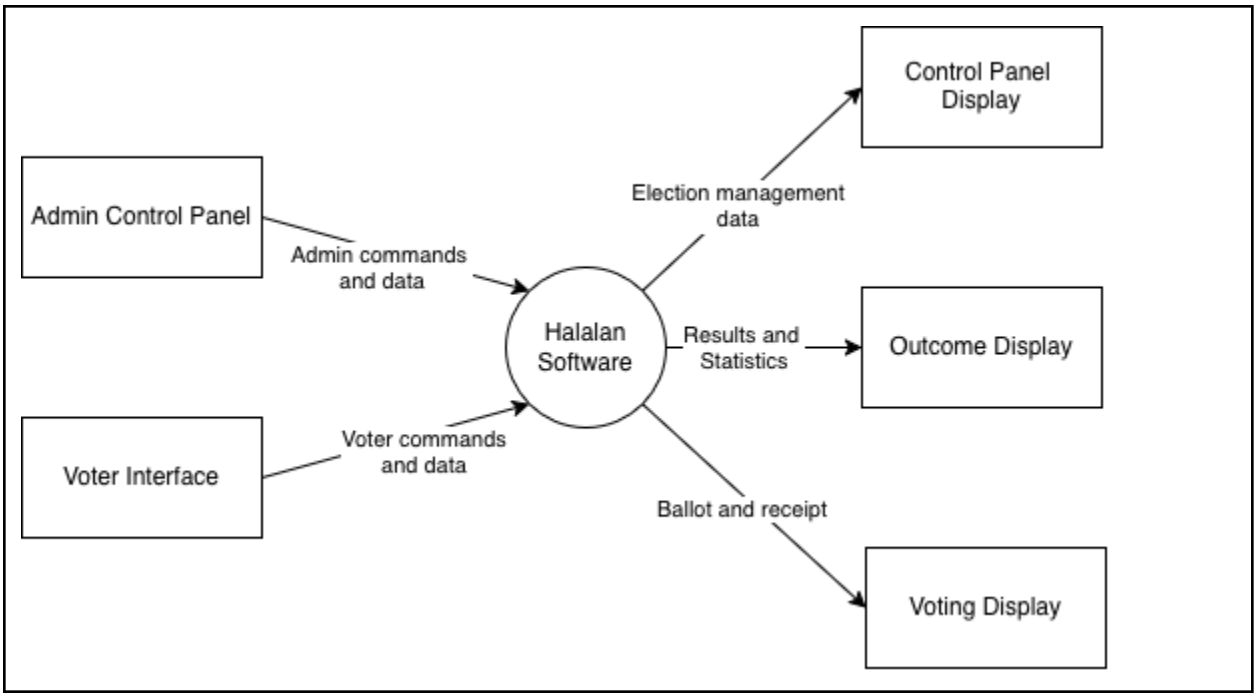


Figure 12. Data Flow Diagram Level 0

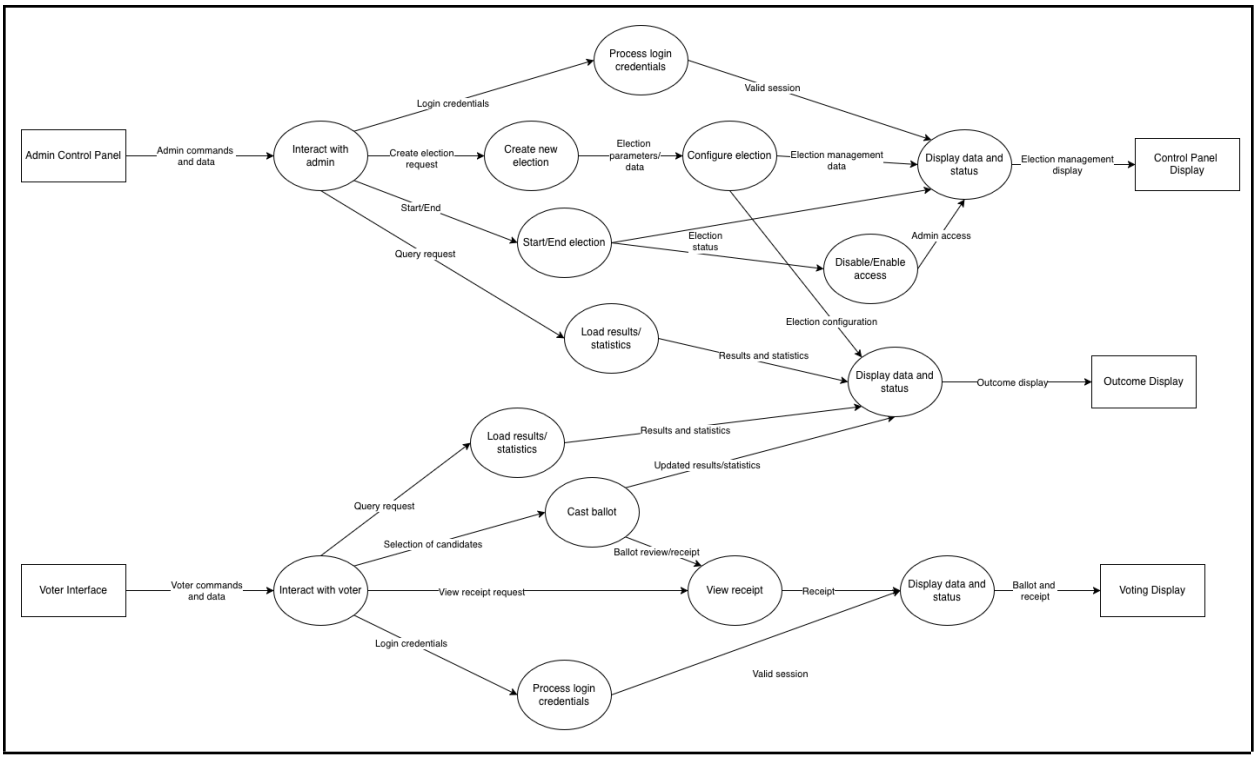


Figure 13. Data Flow Diagram Level 1

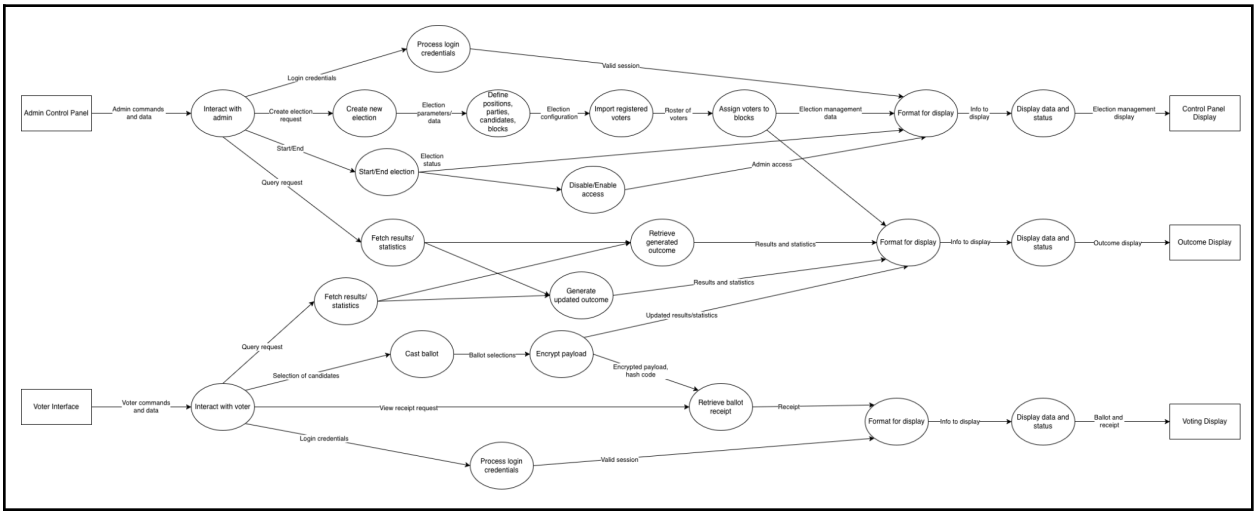


Figure 14. Data Flow Diagram Level 2

2.3 Component-level design

The components are shown below:

Component	Processing	Interfaces	Dependency Type
Credentials Manager	Checks the person against the stored voter or admin credentials.	voter_login(), admin_login(), logout()	Flow
Ballot Handler	Finds the voter's block, and loads the available choices that the voter is eligible for.	vote()	Sharing
Vote Recorder	Prompts the voter user for confirmation of their vote. On confirmation saves the votes, abstains and flags that the user has voted for no double counting	do_vote(), do_verify()	Flow, Constrained
Ballot Receipt Viewer	Collects the voter's own saved choices and displays it to the voter.	votes(election_id)	Flow
Election Manager	Creates, edits and deletes elections, positions, parties, candidates and blocks, and inserts the admin for that election. It rejects a change when an election is running.	insert() update() delete() insert_admin() (on Election object)	Sharing

Voter Users Manager	Adds, edits and deletes voters and lists them by block.	insert() update() delete() (on Voter object)	Sharing
Admin Users Manager	Lists the election administrators and generates a new password or key for each on request.	get_admins(), update(), update_key()	Sharing
Results Counter	Counts votes per candidate and abstains per position and also how many voters registered and voted.	count_votes() count_abstains() count_voters()	Sharing
Activity Logger	Records user activity, and lists the entries for the Logs tab.	insert_log() get_logs()	Sharing

COTS

Component	Purpose
CodeIgniter 4	PHP Framework
MySQL	Selected DBMS
Google OAuth2 package	Verifying a voter's UP Mail through Google sign-in

Algorithms

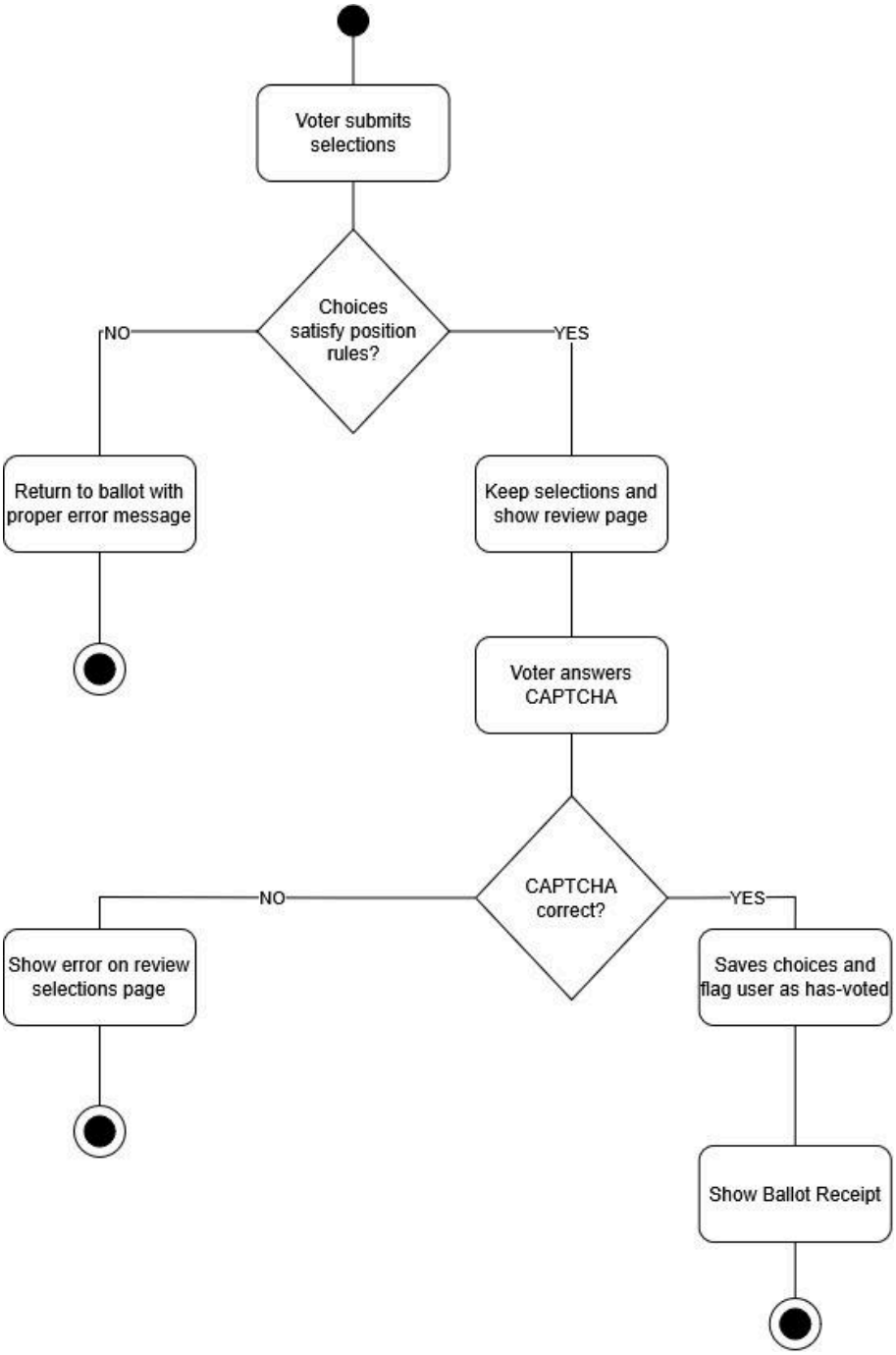


Figure 14. Algorithm

Data Structures

A nested key-value array holds the voter's selections (election, then position, then chosen candidates or abstain), a keyed array holds each position's maximum votes for checking the ballot, and simple lists hold the elections a voter has already voted in and the elections shown to them, with parents listed before their children. Elections form a tree, where each child election points to its parent. These were chosen for simplicity first, since the data is small (a campus-scale election), and then for fast lookup by id.

2.4 Interface elements / Usability design

This section describes the user interface of the Halalan revamp. The revamp keeps the flow of the current Halalan (Figures 1-5 in of the Project Plan) but with a cleaner, more modern look. Additional changes will also be made to account for what the requirements call for such as support for mobile devices, optimized admin interface, vote validation, CAPTCHA, and a more efficient results page. (* – means subject to change)

Interface Design - Menus and Links

a. **General Pages.**

Element/s	Layout	Target Page
Header	Displays the logo at the top left section and the tab bar.	–
Footer	Displays the text 'Powered by Halalan UP.'	–
Tab Bar	Horizontal tab bar heading the page that displays login/logged in as [name] and logout. Current tab open is marked with an underline.	Login and Logout Page
Banners	Rounded alert banners for a more	–

	modern look. Use of green banners for success messages and red banners for errors or notices.	

b. Voter Pages.

Element/s	Layout	Target Page
Login Page	One central maroon button with the text ‘Sign in with your UP Mail.’ Other login methods for users outside of the university will appear as buttons below it.	UP Mail SSO
Ballot Page	One card per position, with each row signifying a candidate (name, party, photo*) and the last row being the abstain option if valid. Lastly, a submit button will be present at the bottom of the page.	Review and Confirm Page
Review and Confirm Page	A summary of all the chosen candidates per position, with Confirm and Back buttons at the bottom.	Receipt or back to Ballot Page

c. **Admin Pages.**

Element/s	Layout	Target Page
Modules Menu	Vertical menu listing HOME, ELECTIONS, POSITIONS, PARTIES, CANDIDATES, BLOCKS, VOTERS, ADMINS, LOGS, with the active module highlighted in maroon	Module List Pages
Home Menu	‘What do you want to do?’* card	Module List Pages,

	with a list of links containing the module list pages, password regeneration, and database reset	Password Regeneration Page, Database Reset
Password Regeneration Page	A textbox for the username, a checkbox for ‘reset login time’, and a regenerate password button. After clicking the button, a feedback banner must appear to signify error or success.	
Database Reset Page	A button for database reset or delete with a confirmation pop-up. After clicking the button, a feedback banner must appear to signify error or success.	
Module List Pages	A table with the election information of the current module, with an add item option at the bottom.	Row Actions
Row actions	Pencil icon to edit, red cross to delete, 'stop' and 'hide' links next to the election status and results state. Order of rows can be modified using drag-and-drop.	–

d. Public Statistics Page

Element/s	Layout	Target Page
Statistics Page	Show the required statistics of the election in card and list form.	–

Aesthetic Design - Layout and graphics

Aspect	Design*
Layout	The main content will be in between the header and footer, containing centered columns with a comfortable width for reading. Most of the content will be inside rounded rectangular cards with a small header containing its title. The layout stays the same across all pages. Lastly, for admin pages, a vertical sidebar on the left is included for the modules menu.
Color	Light-gray background, white cards with black text. Black is used for headers and footers, with light-gray text. UP maroon for buttons and active menu items. Soft red and soft green will be used for success and error cards.
Text	A clean and authoritative sans-serif font will be used (like Helvetica or Inter) to prevent confusion among characters. A readable font size for phones such as 16px will be used for body text, with a proper size difference for headings. **Bold** will be used for headings and position names.
Graphics	The Halalan logo, an updated action set for table managers (add, delete, info, confirm), and candidate photos (TBD).

Navigation Design

Given Halalan is a voting website, navigation between pages must be clear and easy to understand. On the other hand, it must not be too easy to click important actions such as submitting a ballot, or creating an election. Therefore, there must exist a balance between user-friendliness and ensuring safety protocols when taking into account the navigation design of these pages.

With the table below, typical **navigation scenarios** will be discussed based on two types of actions. Navigation actions are clicks on links, tabs and buttons that move between pages or submit a page. Data entry actions are keystrokes and selections that supply content: typing, choosing a dropdown value, ticking a candidate, dropping a file.

Scenario	Navigation Actions	Data Entry Actions	Total
1a. Voter votes, signing in to UP Mail for the first time	7	35~	42~
1b. Voter votes, already signed in UP Mail	5	5	8
2. Admin adds one candidate	3	23~	26~
3. Admin imports a voter list	4	3	7
4, Public viewer reads the turnout statistics	1	0	1

These are the steps behind each scenario:

1. Voter votes (Given five positions, with one choice each). Open the Halalan link (1N), click Sign in with your UP Mail (1N), click Next (1N) and Sign in (1N) on the UP Mail page, type your UP mail and your password (~30D), tick one candidate in each of the five positions (1D), click Submit Ballot (1N), click Confirm (1N), and click Log out (1N).
    a. For scenario 1a, there is a total of 7N and an approximated 33D which can vary depending on the length of the user email and password. Thus, 16.67% of its actions are navigation actions.
    b. Scenario 1b skips the typing of the UP mail and the password, as well as the two clicks on the UP Mail page, which leaves 5N and 3D. Thus, 50% of its actions are navigation actions.

2. 2. Admin adds a candidate. Click CANDIDATES (1N), click Add Candidate (1N), choose the election, position and party from dropdowns (6D), type a first and last name (~16D), drag in a photo (1D), and click Save (1N). This results in 3N and 23D. Thus, 11.5% of its actions are navigation actions.
3. Admin imports a voter list. Click VOTERS (1N), click Import voters (1N), choose the election (2D), drop the CSV file (1D), click Upload (1N), and click Confirm on the summary (1N). Thus, 57.14% of its actions are navigation actions.
4. Public viewer reads the turnout. Open the results link (1N). Thus, 100% of its actions are navigation actions.

Next, **navigation semantic units (NSUs)** will be studied on each page based on the ways of navigating them that together complete one user goal.

NSU	Used by	Ways of Navigation
Authenticate	Voter	Login page, UP mail SSO, CAPTCHA
Cast Ballot	Voter	Ballot page, Review and confirm page, Receipt
Set-up election	Admin	Admin home page, ELECTIONS, POSITIONS, PARTIES, CANDIDATES, BLOCKS (each tab with its own add and edit forms)
Manage voters	Admin	Voter list, Import form, Import summary, Regeneration Box
Run election	Admin	Elections page, start or stop
View results	All	Results page, statistics page

References

Pressman, R. S., & Maxim, B. R. (2010). *Software engineering: A practitioner's approach* (7th ed.). R.S. Pressman & Associates, Inc.